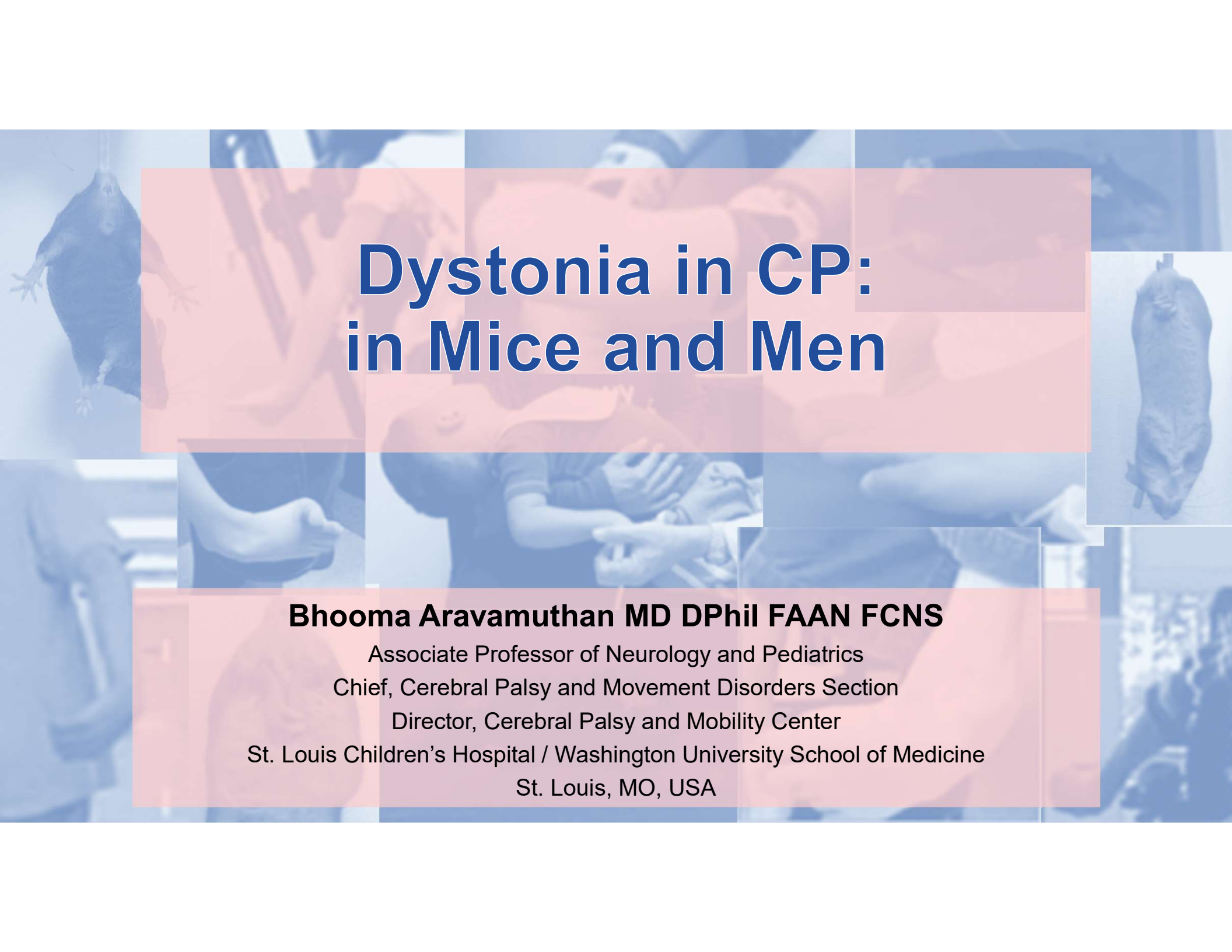


The background is a collage of images. On the left and right sides, there are images of mice, one of which appears to be hanging. In the center, there are images of children, including one who is lying down and another who is sitting up. The entire background is overlaid with a semi-transparent red rectangle that contains the title and the speaker's information.

Dystonia in CP: in Mice and Men

Bhooma Aravamuthan MD DPhil FAAN FCNS

Associate Professor of Neurology and Pediatrics

Chief, Cerebral Palsy and Movement Disorders Section

Director, Cerebral Palsy and Mobility Center

St. Louis Children's Hospital / Washington University School of Medicine

St. Louis, MO, USA

Disclosures

- None relevant
- Research funding from:
 - National Institute of Neurological Disorders and Stroke
 - National Institute of Child Health and Human Development
 - National Center for Medical Rehabilitation Research Pediatric Epilepsy Research Foundation
 - Pediatric Epilepsy Research Foundation
 - Child Neurology Foundation
 - Centers for Disease Control
 - American Board of Psychiatry and Neurology
 - Rappaport Family Foundation
 - CP Family Fund
- Royalties from UpToDate, Inc.



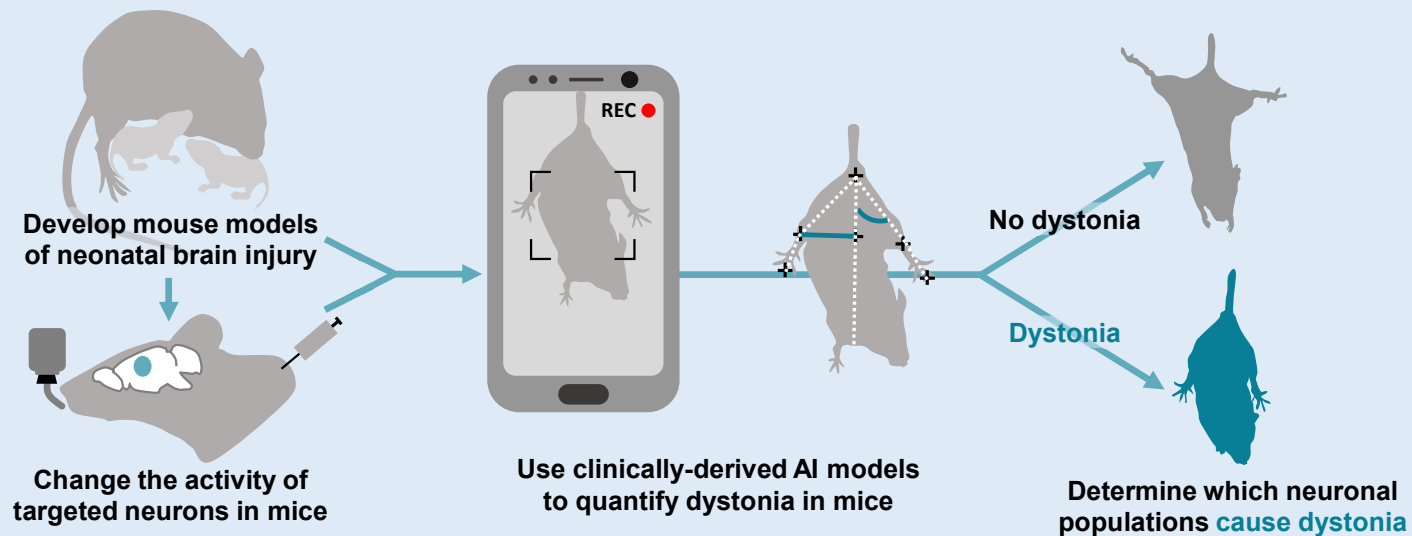
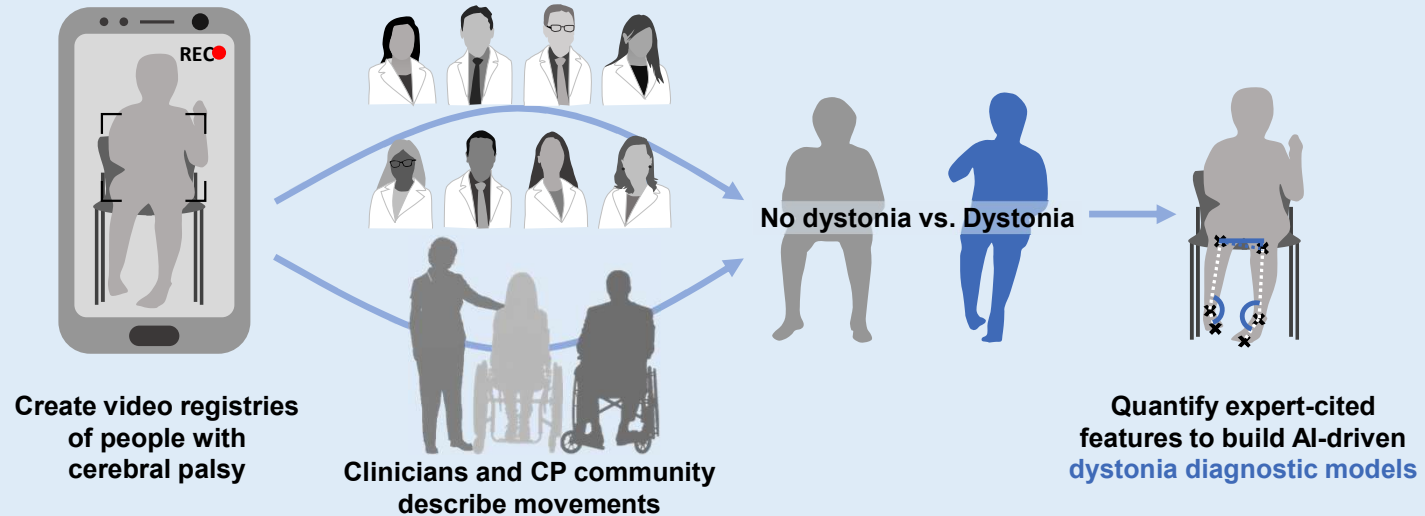
Our goal is to cure
dystonia in people
with **cerebral palsy**

Washington University School of Medicine in St. Louis

Children's
HOSPITAL • ST. LOUIS

 aravamuthanlab.wustl.edu

 aravamuthanb@wustl.edu



If measures that translate from
to do not exist,



Gemperli...Aravamuthan, *Ann Neurol.* 2025

If measures that translate from
to do not exist,
we must **create** them



Gemperi...Aravamuthan, *Ann Neurol.* 2025

Dystonia is:
voluntary movement triggered
overflow muscle activation;
arousal-dependent

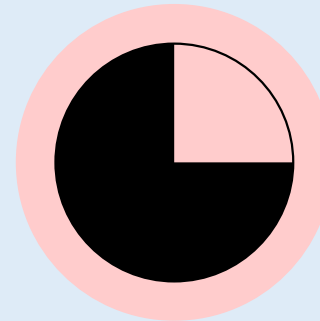
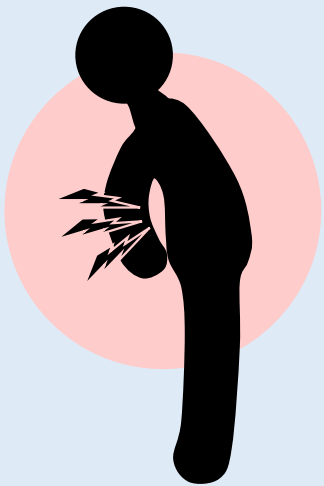
Albanese et al., *Mov. Dis.* 2025; Sanger et al., *Pediatrics*, 2003

Dystonia is:



Albanese et al., *Mov. Dis.* 2007; Sanger et al., *Pediatrics*, 2003; Lin et al., *JNNP* 2014; Perides et al. *DMCN* 2020; Yuan-Kim Liow et al, *Eur. J. Pediatr. Neurol.* 2016; Monbaliu et al., *DMCN* 2017; Fehlings et al. *DMCN* 2018; Knights et al., *J. Child Neurol.* 2013; Rice et al. *DMCN* 2017; Aravamuthan et al. *DMCN* 2021

Dystonia is:



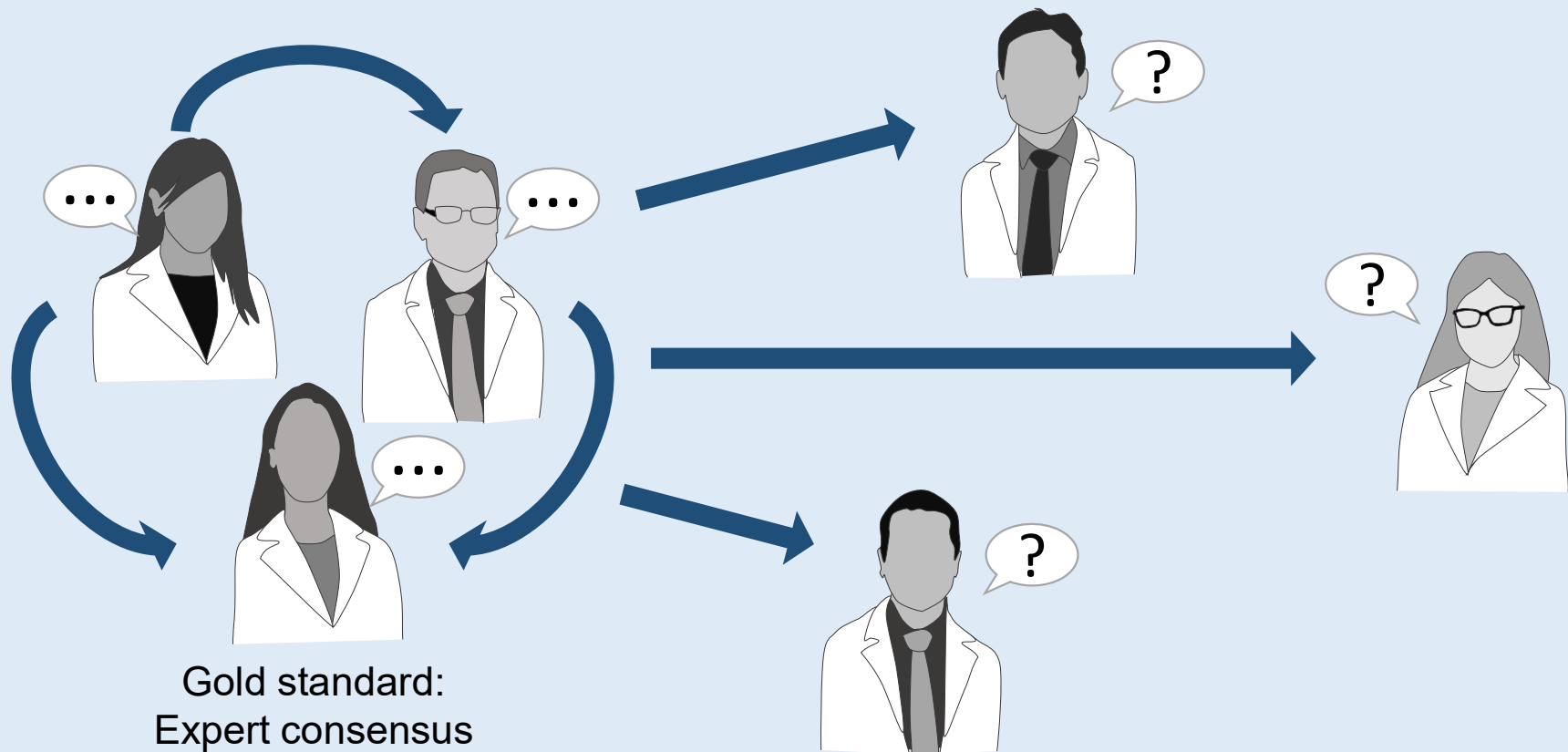
12%

Albanese et al., *Mov. Dis.* 2007; Sanger et al., *Pediatrics*, 2003; Lin et al., *JNNP* 2014; Perides et al. *DMCN* 2020; Yuan-Kim Liow et al, *Eur. J. Pediatr. Neurol.* 2016; Monbaliu et al., *DMCN* 2017; Fehlings et al. *DMCN* 2018; Knights et al., *J. Child Neurol.* 2013; Rice et al. *DMCN* 2017; Aravamuthan et al. *DMCN* 2021

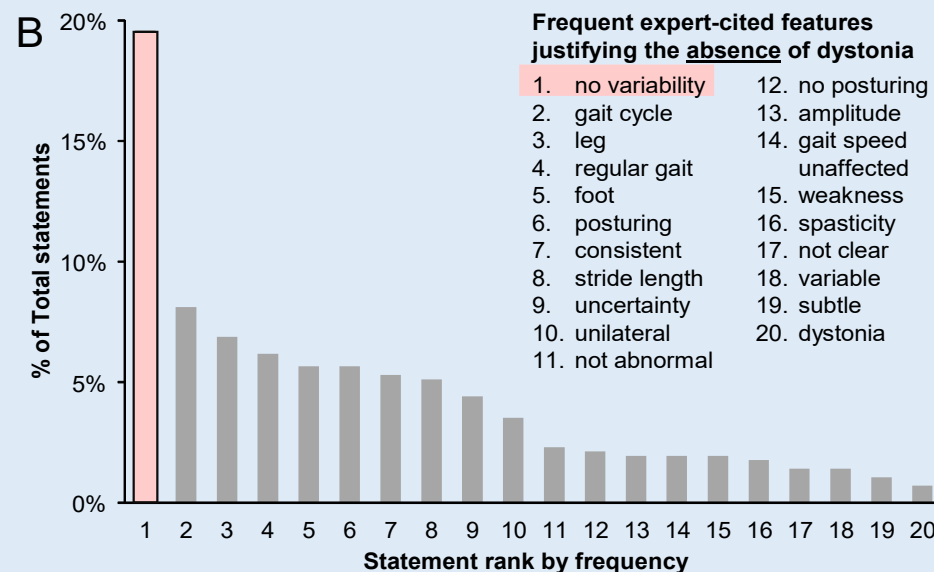
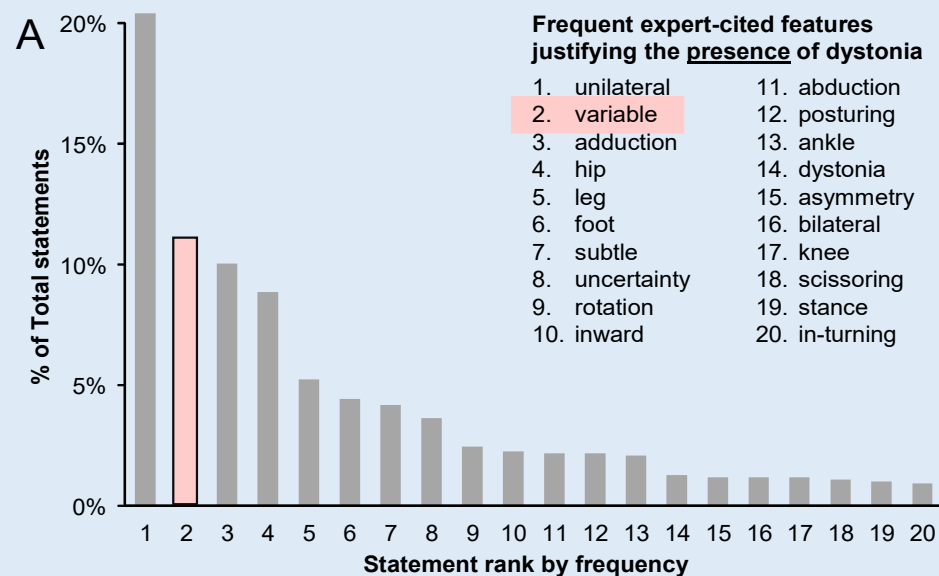
Dystonia is: Leg adduction variability and amplitude

Aravamuthan et al. *DMCN* 2021; Aravamuthan et al. *DMCN* 2022; Gemperli...Aravamuthan. *Ann Neurol*. 2025

Dystonia diagnosis: status quo

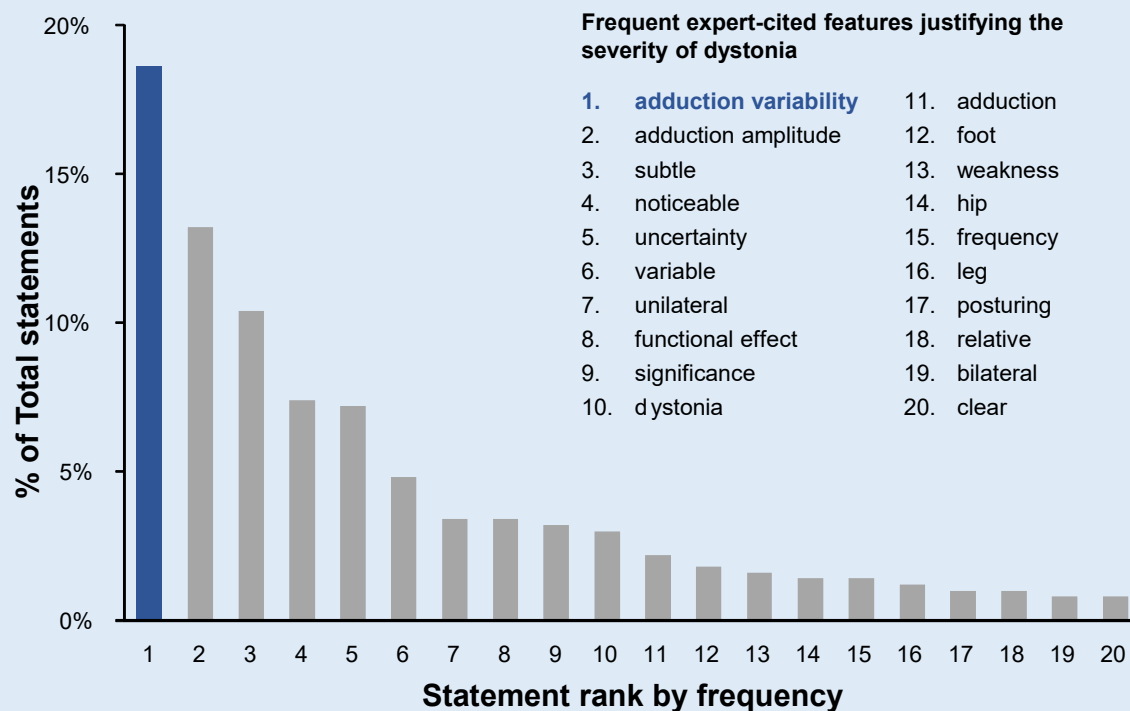


Expert-cited diagnostic features



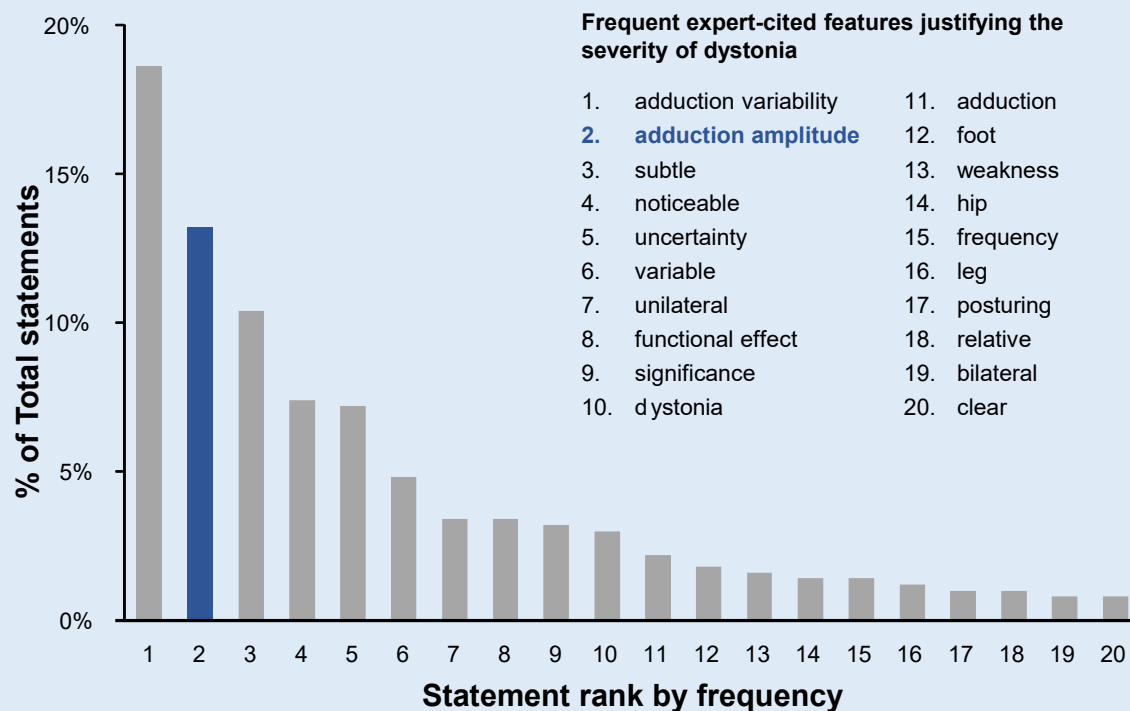
Aravamathan et al., *Dev Med Child Neurol.* 2021; *Dev Med Child Neurol.* 2023

Expert-cited severity features



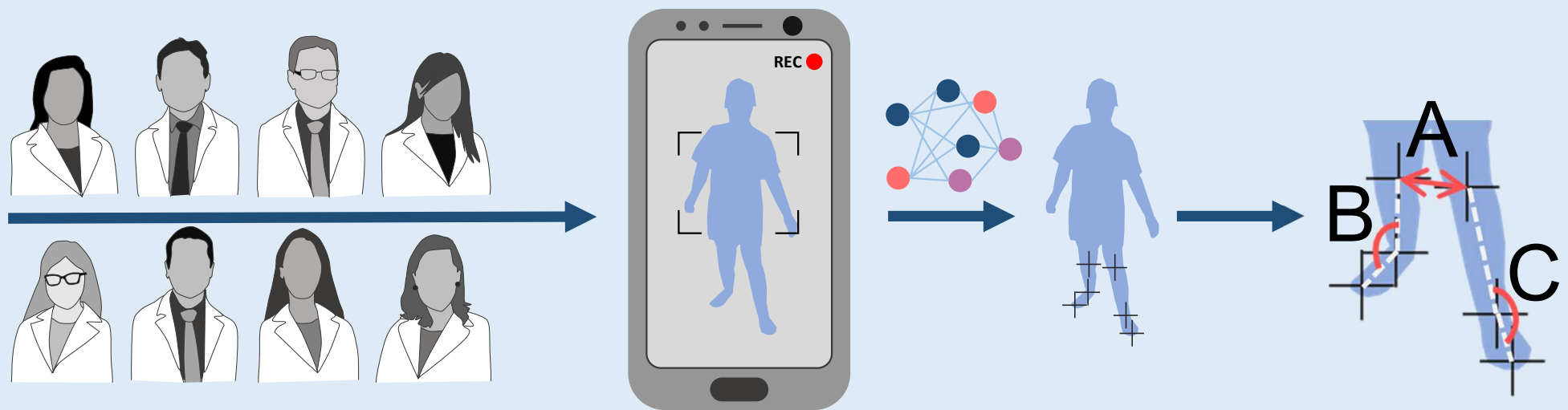
Aravamathan et al., *Dev Med Child Neurol.* 2021; *Dev Med Child Neurol.* 2023

Expert-cited severity features



Aravamathan et al., *Dev Med Child Neurol.* 2021; *Dev Med Child Neurol.* 2023

Dystonia assessment: a new way

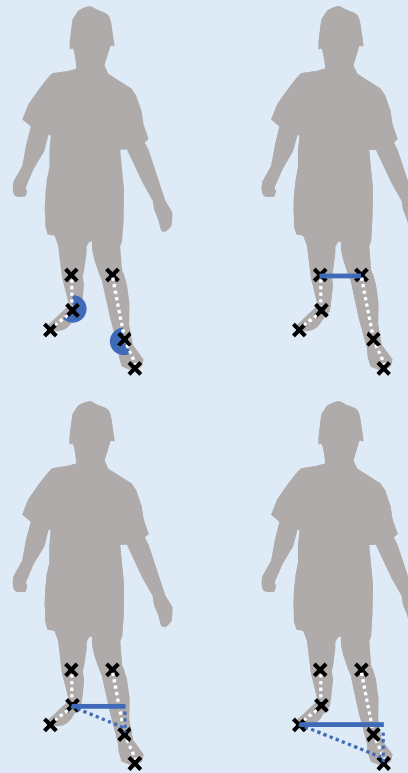
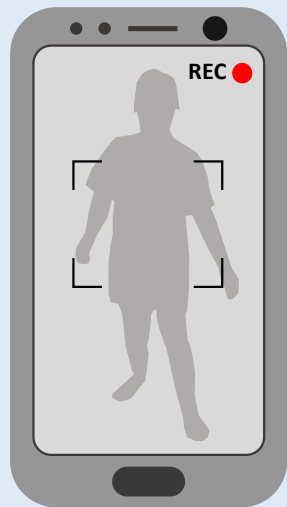


What are the expert-cited features of dystonia?

Develop gait variable measures of these expert-cited features

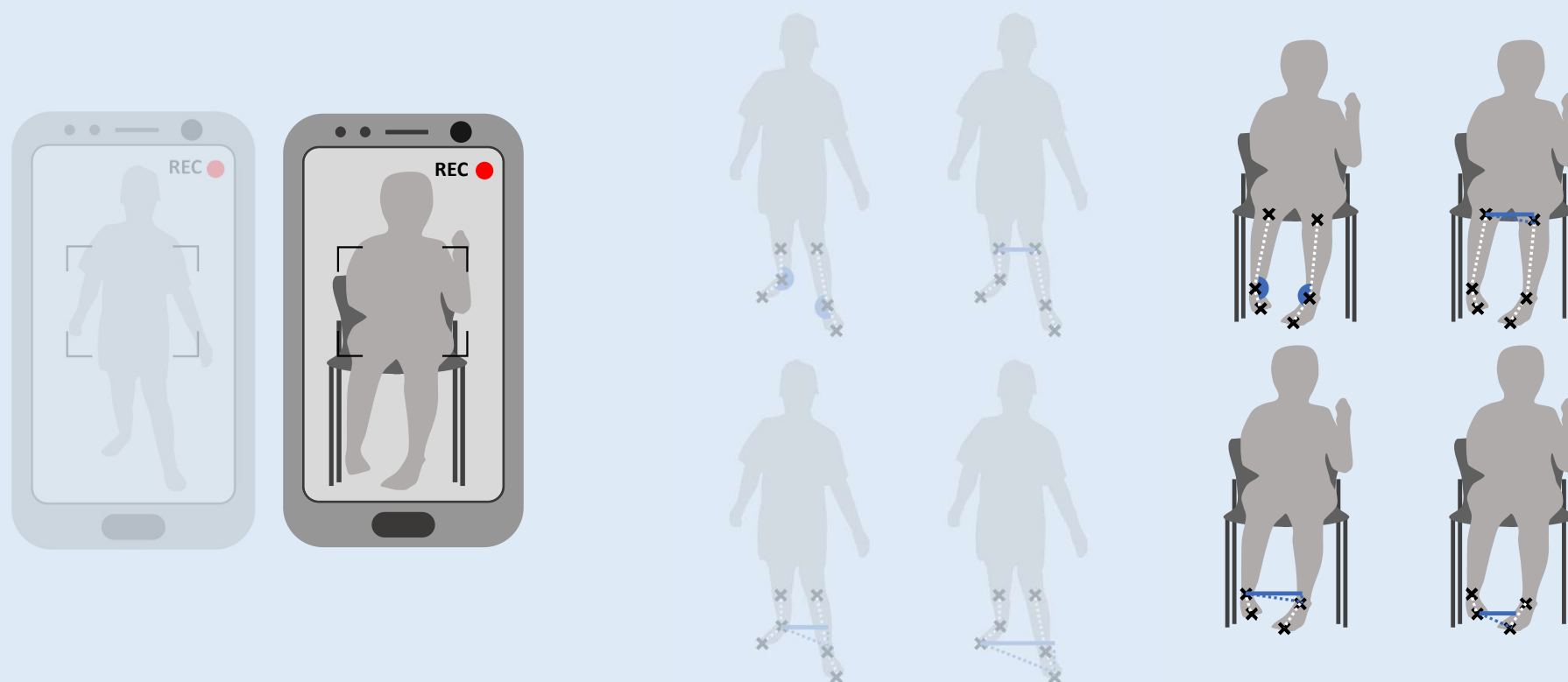
Aravamuthan et al., *Dev Med Child Neurol.* 2021; *Dev Med Child Neurol.* 2023

Quantifying dystonia severity



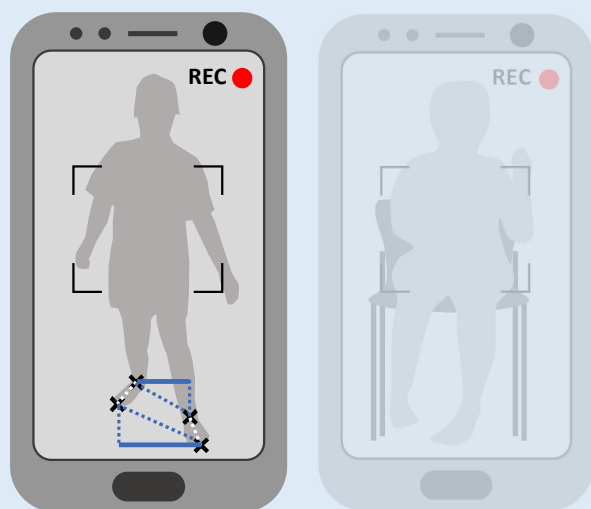
Aravamuthan et al. *Dev Med Child Neurol* 2023; Gemperli...Aravamuthan *Ann Neurol* 2025

Quantifying dystonia severity

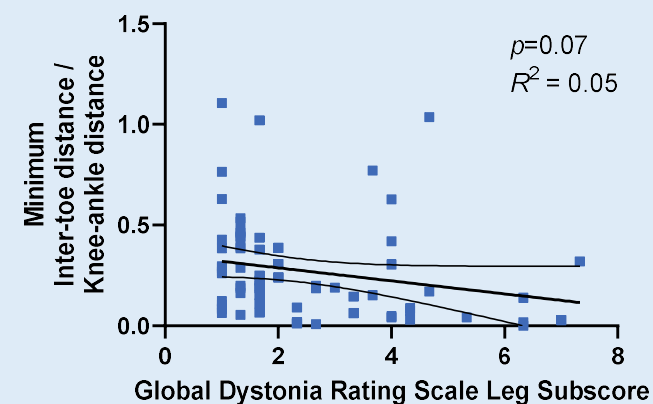
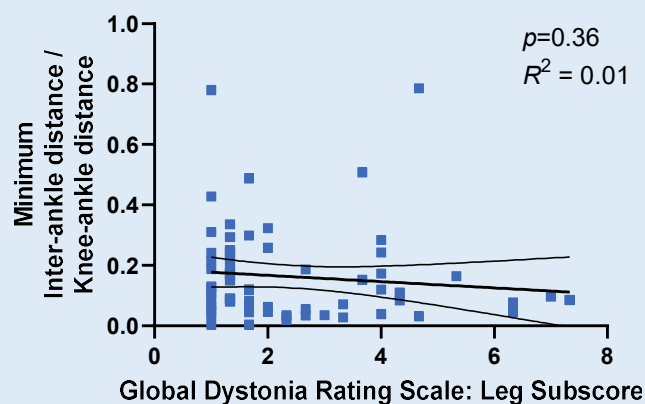


Aravamuthan et al. *Dev Med Child Neurol* 2023; Gemperli...Aravamuthan *Ann Neurol* 2025

Quantifying dystonia severity



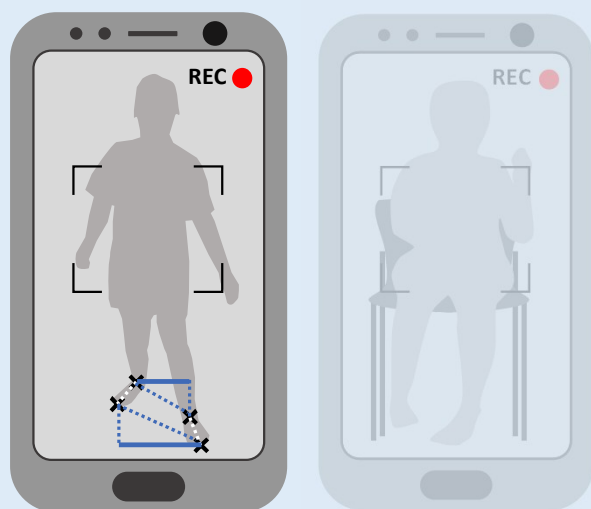
Lower limb adduction amplitude



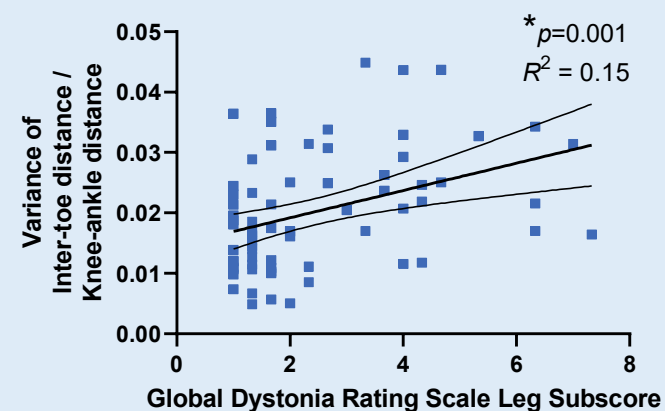
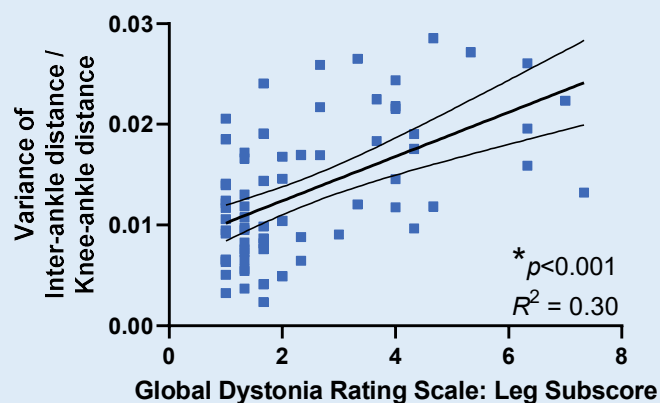
*Multiple linear regression, n=70

Aravamuthan et al. *Dev Med Child Neurol* 2023; Gemperli...Aravamuthan *Ann Neurol* 2025

Quantifying dystonia severity



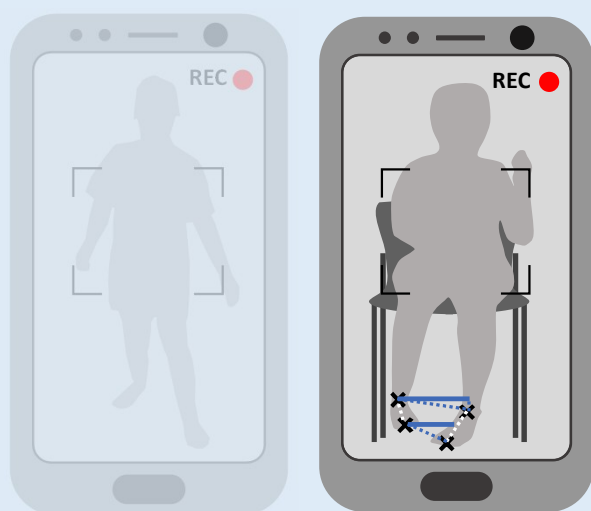
Lower limb adduction variability



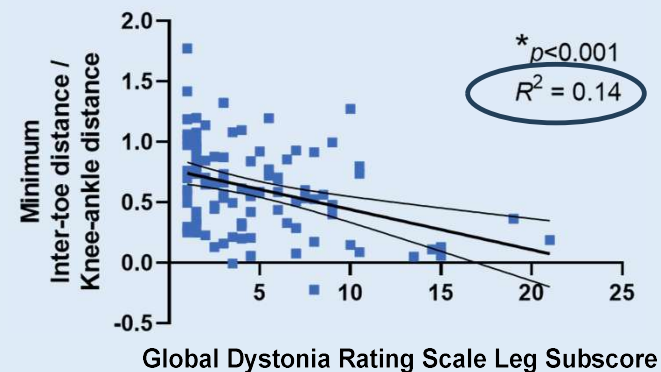
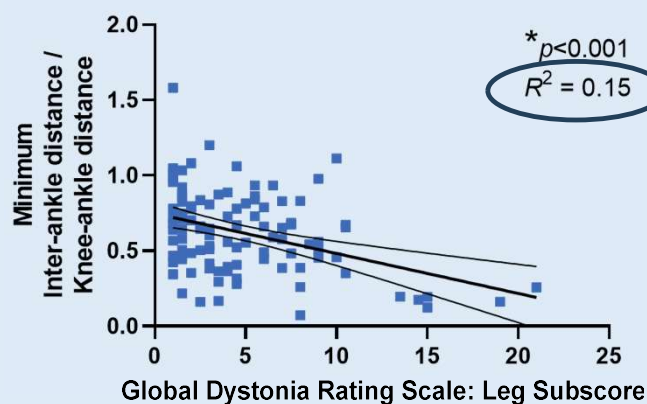
*Multiple linear regression, n=70

Aravamathan et al. *Dev Med Child Neurol* 2023; Gemperli...Aravamathan *Ann Neurol* 2025

Quantifying dystonia severity



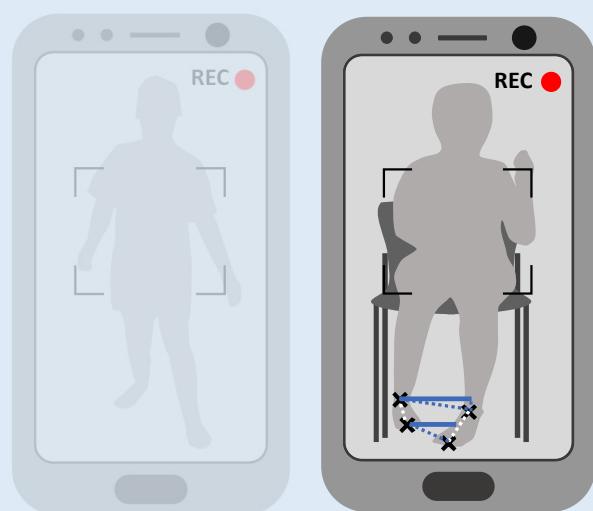
Lower limb adduction amplitude



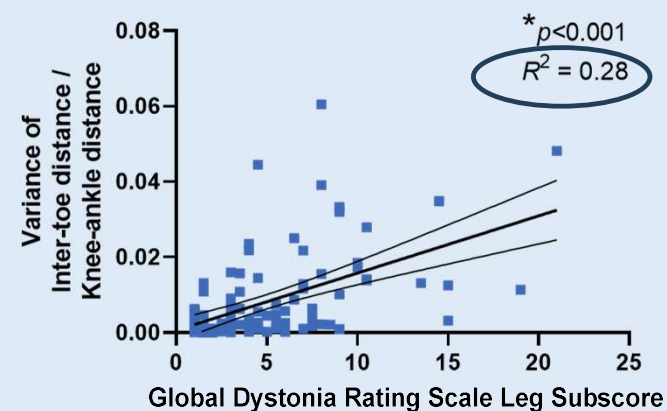
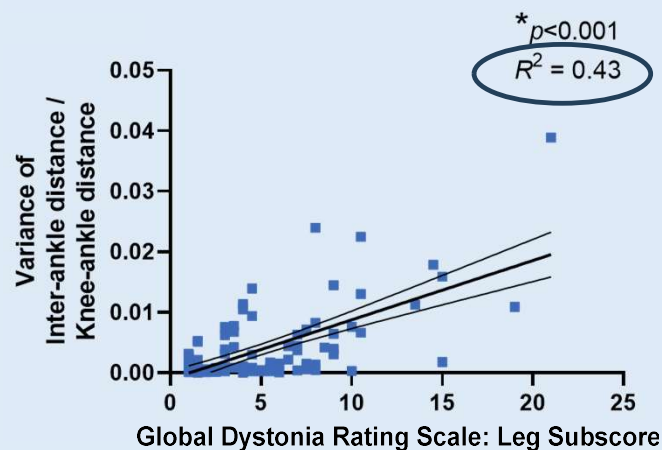
*Multiple linear regression, n=104

Aravamathan et al. *Dev Med Child Neurol* 2023; Gemperli...Aravamathan *Ann Neurol* 2025

Quantifying dystonia severity



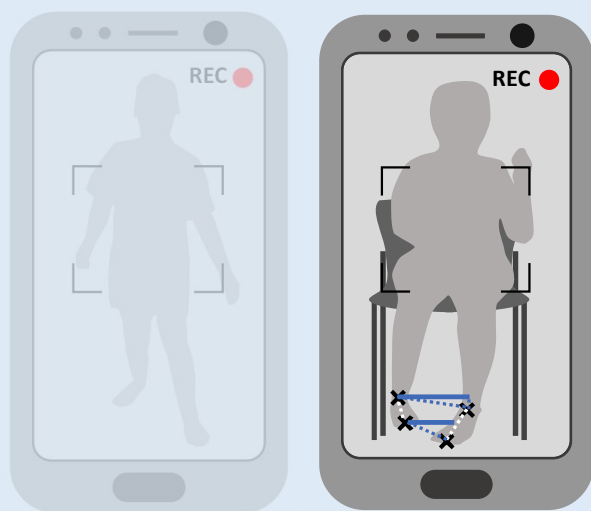
Lower limb adduction variability



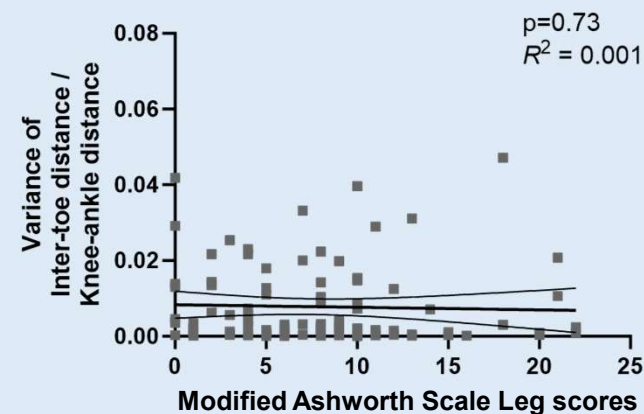
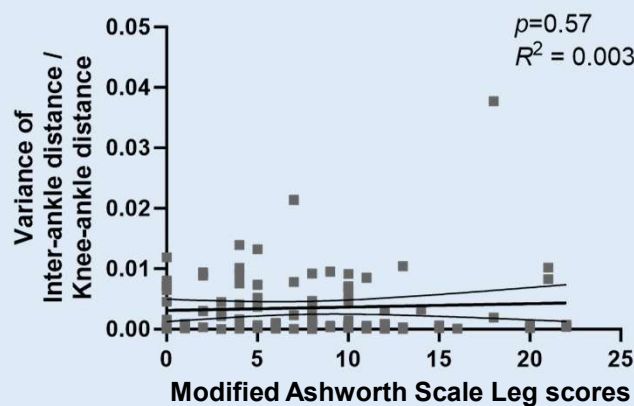
*Multiple linear regression, n=104

Aravamuthan et al. *Dev Med Child Neurol* 2023; Gemperli...Aravamuthan *Ann Neurol* 2025

Quantifying dystonia severity, **not** spasticity



Lower limb adduction variability



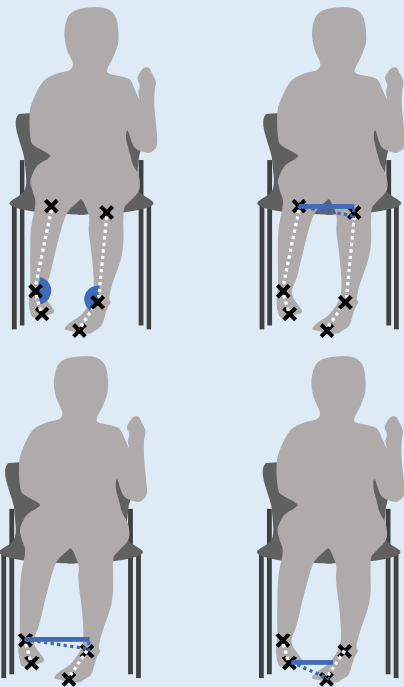
*Multiple linear regression, n=89

Aravamuthan et al. *Dev Med Child Neurol* 2023; Gemperli...Aravamuthan *Ann Neurol* 2025

Machine learning for dystonia assessment



Rishabh Bajpai



DxTonia:

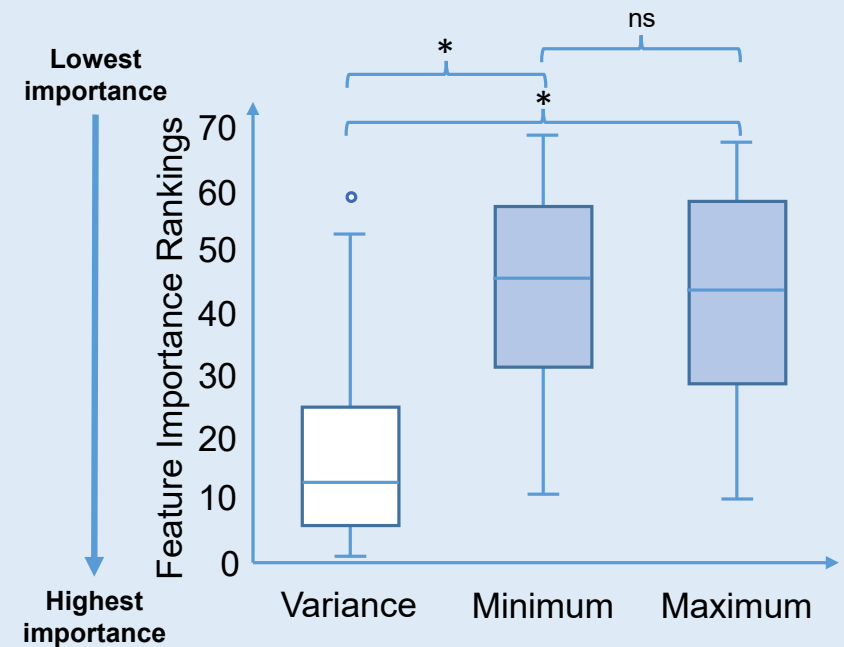
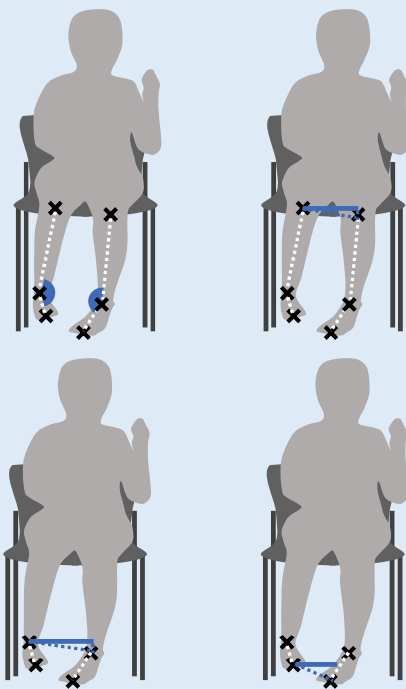
Open-source software
to identify leg dystonia
in videos of people with CP

Bajpai...Aravamuthan. *Ann Neurol.* 2025

Machine learning for dystonia assessment



Rishabh Bajpai

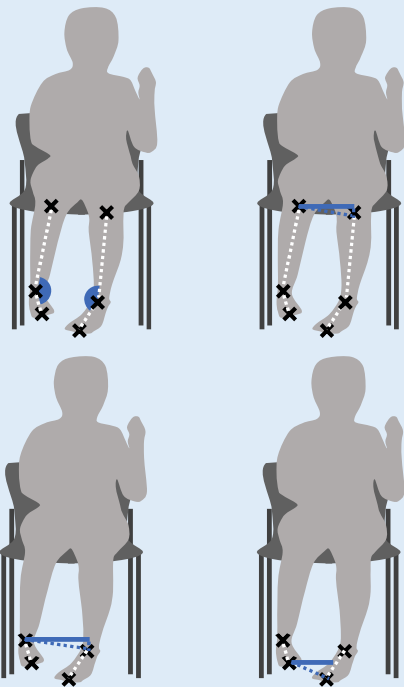


Bajpai...Aravamuthan. *Ann Neurol.* 2025

Machine learning for dystonia assessment



Rishabh Bajpai



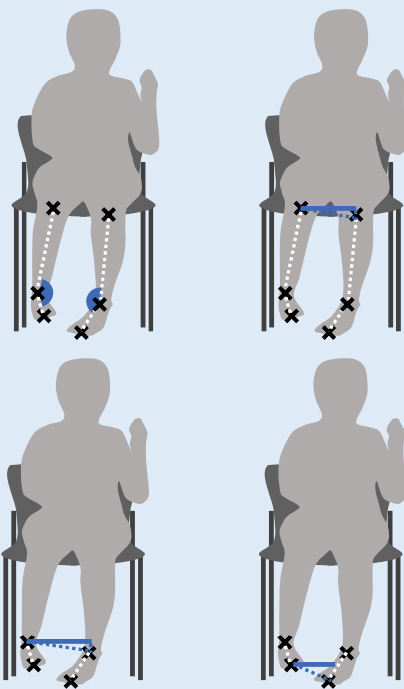
DxTonia:
83% accuracy
(compared to expert consensus)

Bajpai...Aravamuthan. *Ann Neurol.* 2025

Machine learning for dystonia assessment



Rishabh Bajpai

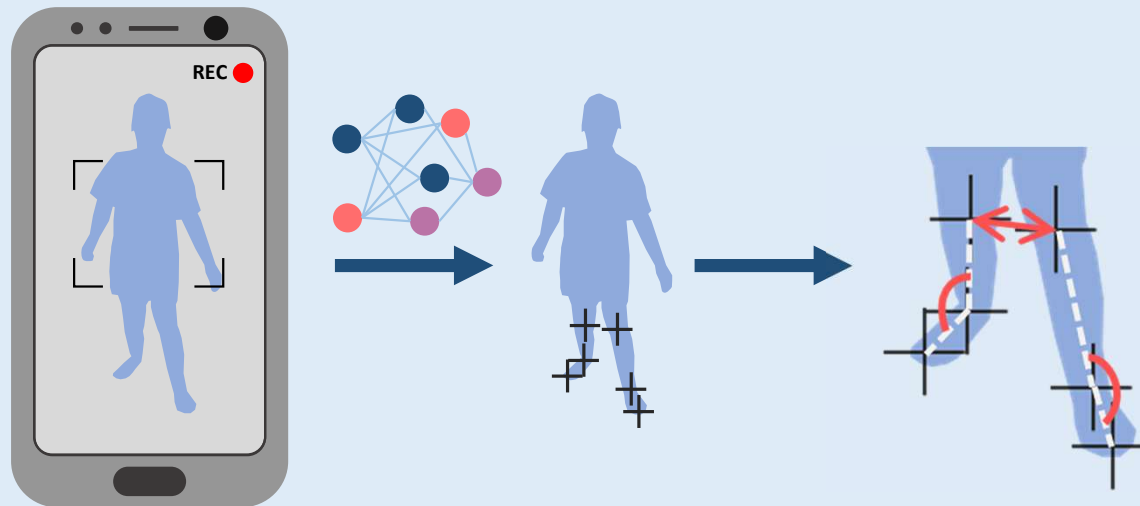


DxTonia:

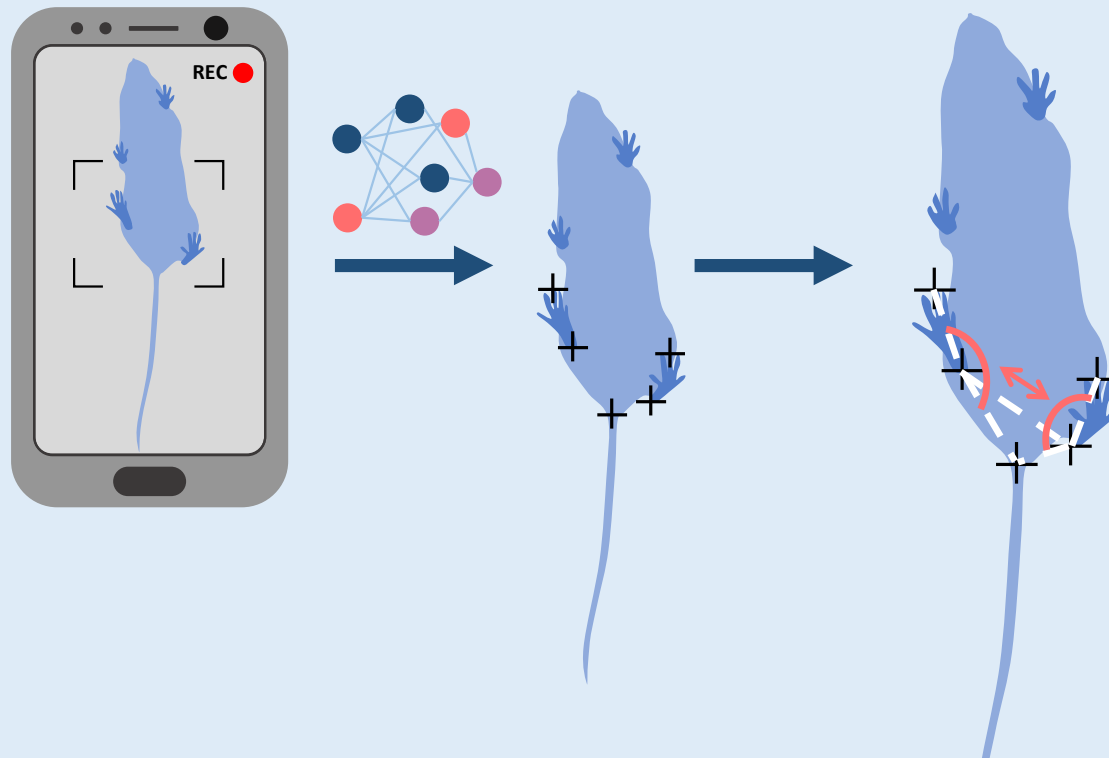
90% accuracy in estimating
dystonia severity (GDRS)

Bajpai...Aravamuthan. *Ann Neurol.* 2025

Dystonia assessment: a new way



Dystonia assessment: a new way



Dystonia diagnosis: status quo



Sam Pappas

Bill Dauer

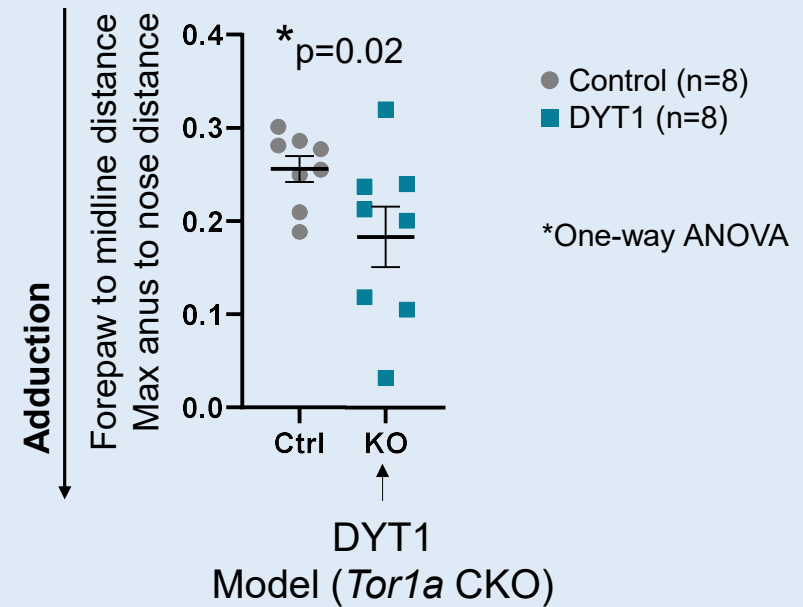
Wild-type



Model of DYT1 (*Tor1a* forebrain knockout)

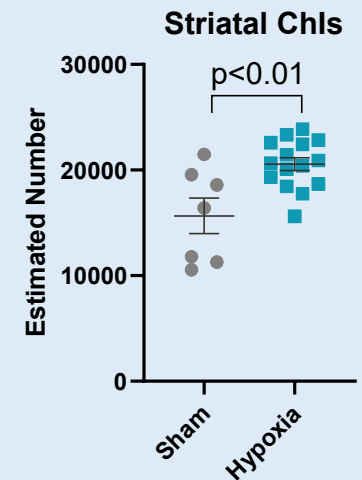
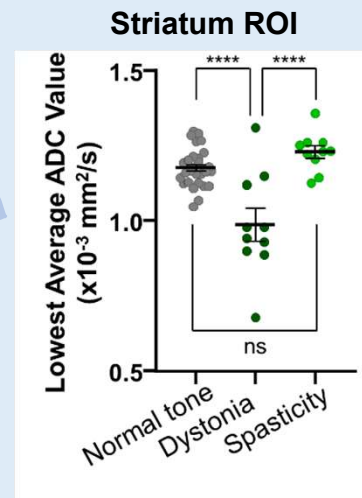
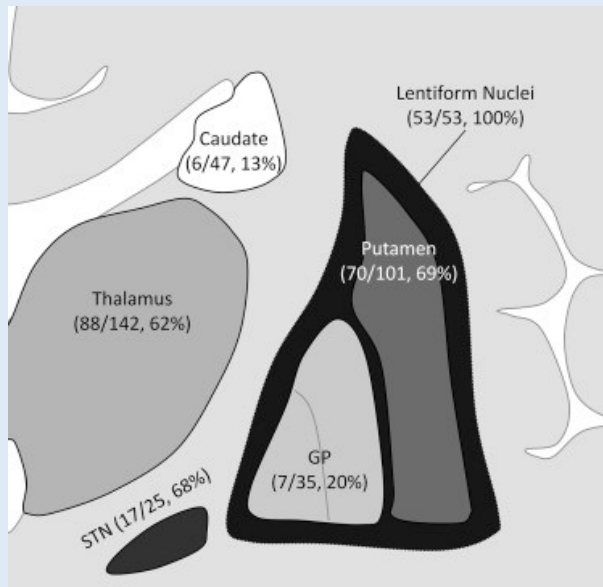


Hindpaw Distance from Midline (Adduction)



Sanger and Ferman., *Mov Disord Clin Pract.* 2017; Jinnah et al. *Mov Disord.* 2005; Oleas et al. *Mov Disord* 2013.

Dystonia pathophysiology – striatum



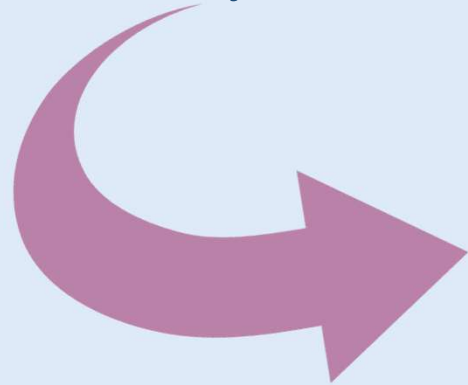
Diverse Mechanisms Lead to Common Dysfunction of Striatal Cholinergic Interneurons in Distinct Genetic Mouse Models of Dystonia

Karen L. Eskow Jaunarajs,^{1*} Mariangela Scarduzio,^{1*} Michelle E. Ehrlich,² Lori L. McMahon,^{1,2} and David G. Standaert¹

Dystonia is **ASSOCIATED** with abnormal excitation of striatal cholinergic interneurons (ChIs)

Aravamuthan and Waugh, *Pediatric Neurology* 2016; Chintalapati...Aravamuthan, *Pediatric Neurology* 2020; Gandham*, Tak*, and Aravamuthan, *Neurobiol Dis.* 2020

Striatal
cholinergic interneuron (ChI)
excess activity / number



Causative?

Dystonia

Injury



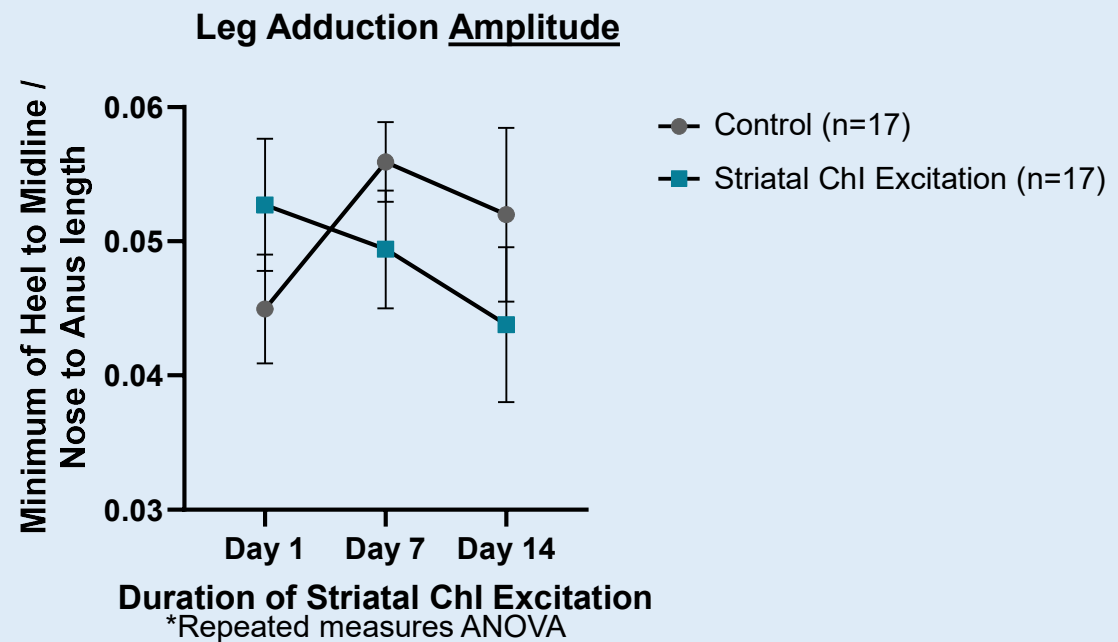
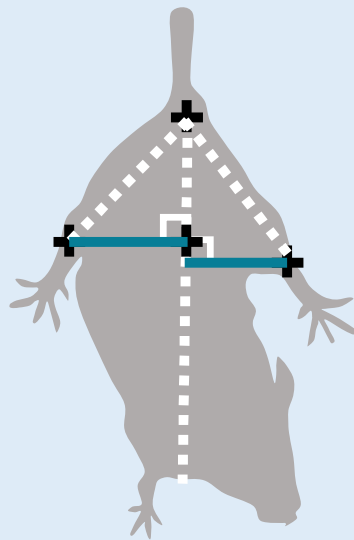
Necessary?

Gandham...Aravamuthan, *Neurobiol Dis.* 2021; Eskow Jaunarajs et al., *Prog Neurobiol.* 2019, *J Neurosci.* 2015; Pisani et al., *Neurobiol Dis.* 2006; Sciamanna et al., *PLoS One* 2011; Lin et al. *J Neurol Neurosurg Psychiatry* 2014

Does striatal ChI excitation cause dystonia?



Jordan McCall

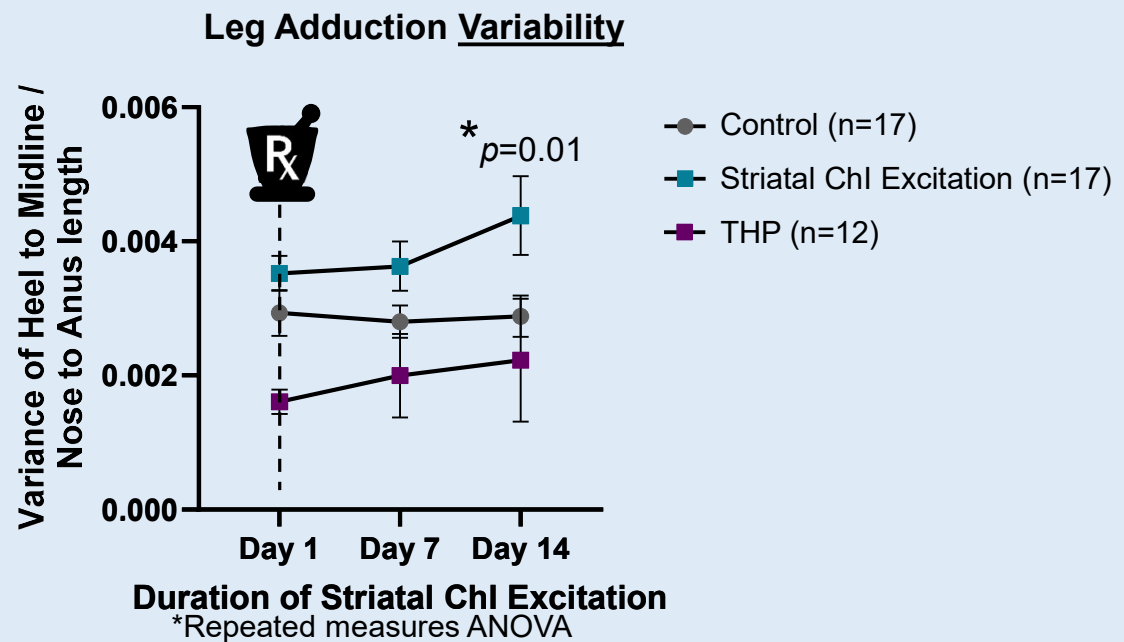
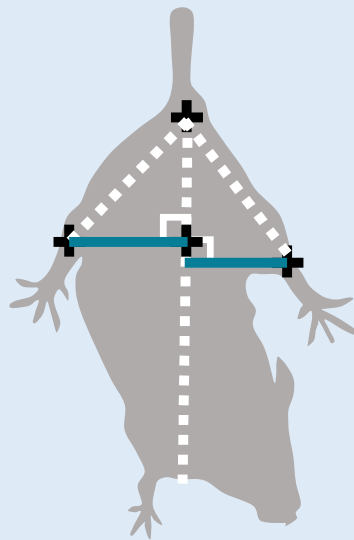


Gemperli...Aravamuthan. *Ann Neurol* 2025

Does striatal ChI excitation cause dystonia?



Jordan McCall



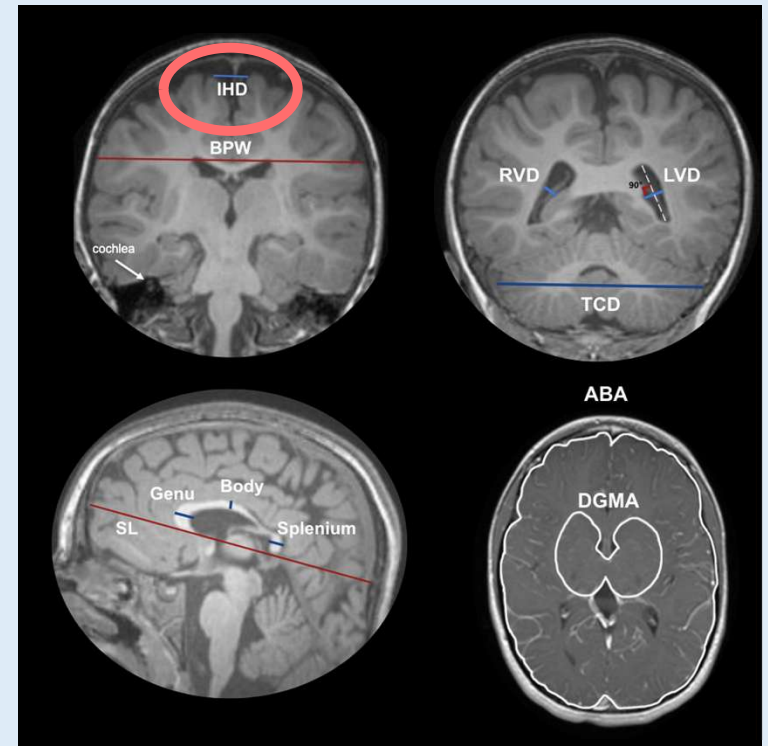
Gemperli...Aravamuthan. *Ann Neurol* 2025

Is the striatum the whole story?

Of people with CP and
isolated PVL on MRI

59%

have dystonia



Ueda, Aravamuthan, and Pearson.
Dev Med Child Neurol 2022

Chintalapati...Aravamuthan. *Pediatr Neurol.* 2023

Sensorimotor cortex and dystonia

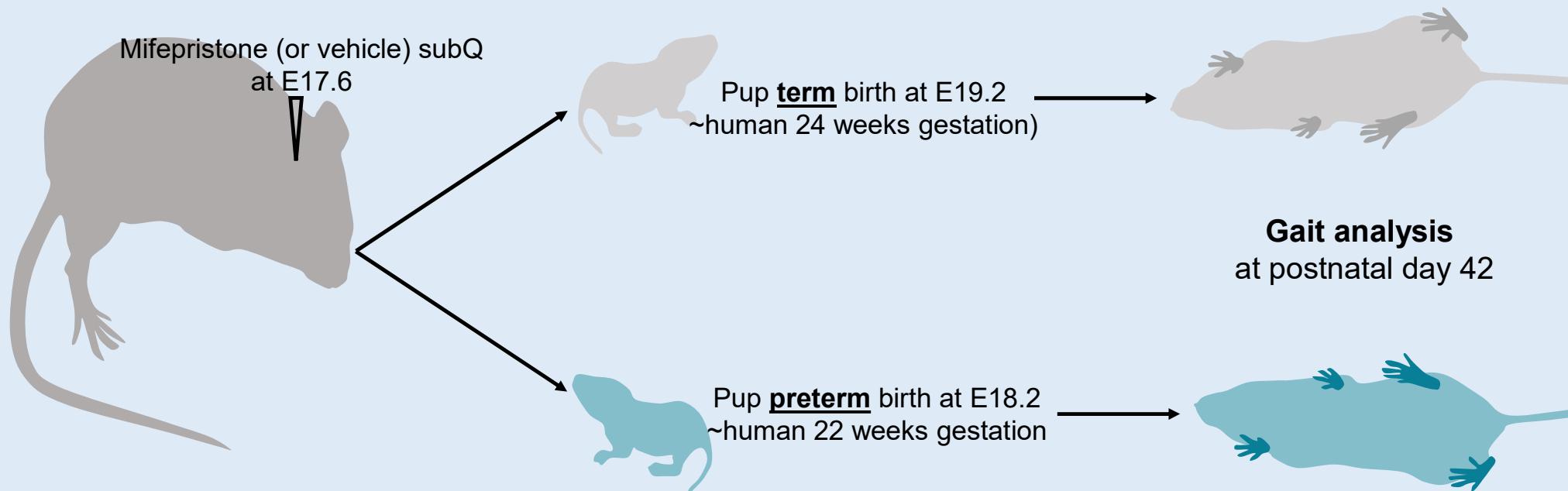


The collage features three journal covers:

- Elsevier NeuroImage: Clinical**: Volume 17, 2018, Pages 90-97. Article: "Loss of inhibition in focal hand dystonia". Authors: Cecile Gallean, Priyan, Ziad Saad, Shantalaxmi Thad.
- Brain**: Article: "Network localization". Authors: Daniel T Corp, Juho Joutsma, R Rya, Cecilia N Prudente, Jianxun Ren, Mar.
- JAMA Neurology**: Article: "Regional, not global, functional connectivity contributes to isolated focal dystonia". Published September 10, 2020. Authors: Scott A. Norris, MD, Aimee E. Morris, PhD, Meghan C. Campbell, PhD, Morvarid Karimi, MD, Babatunde Adeyemo, BA, Randal C. Paniello, MD, PhD, Abraham Z. Snyder, MD, PhD, Steven E. Petersen, PhD, Jonathan W. Mink, MD, PhD, and Joel S. Perlmutter, MD.

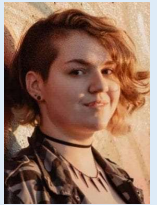
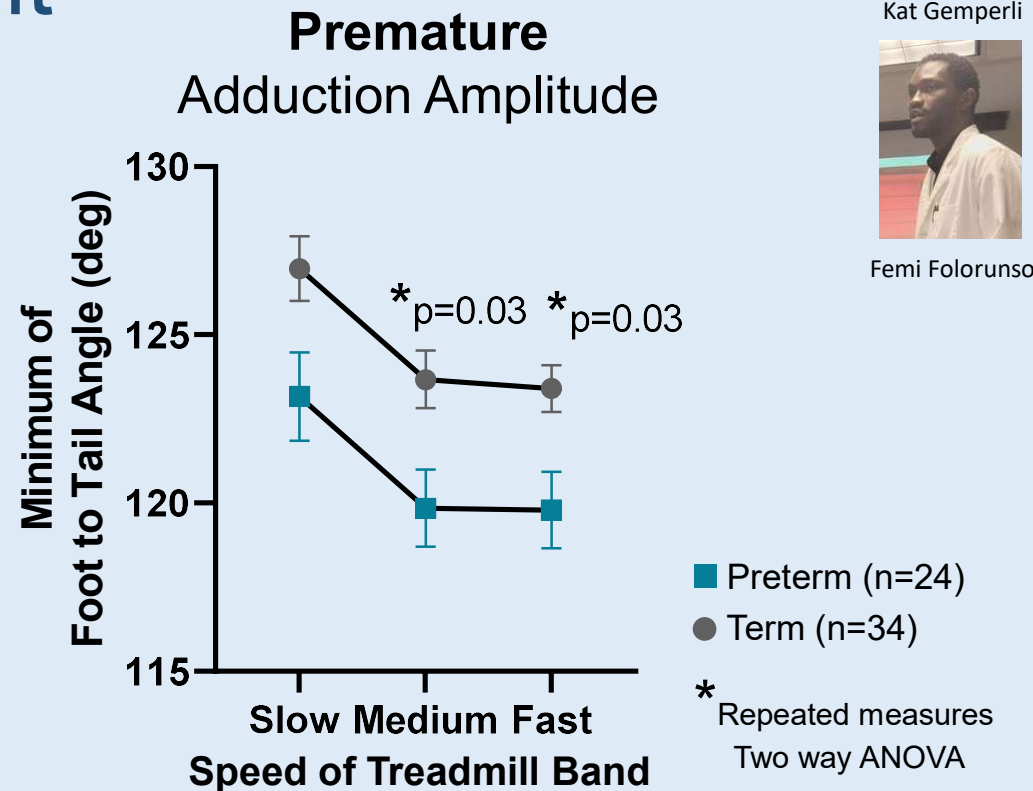
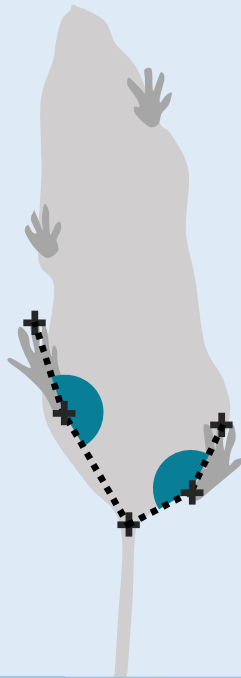
Dystonia is **ASSOCIATED** with abnormal inhibition of sensorimotor cortex

Mouse model of preterm birth



Gemperli...Galindo, Aravamuthan. *Ped Res.* 2023

Assessing clinically-relevant dystonia features in mice - gait

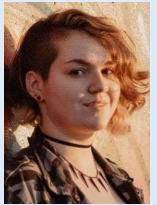
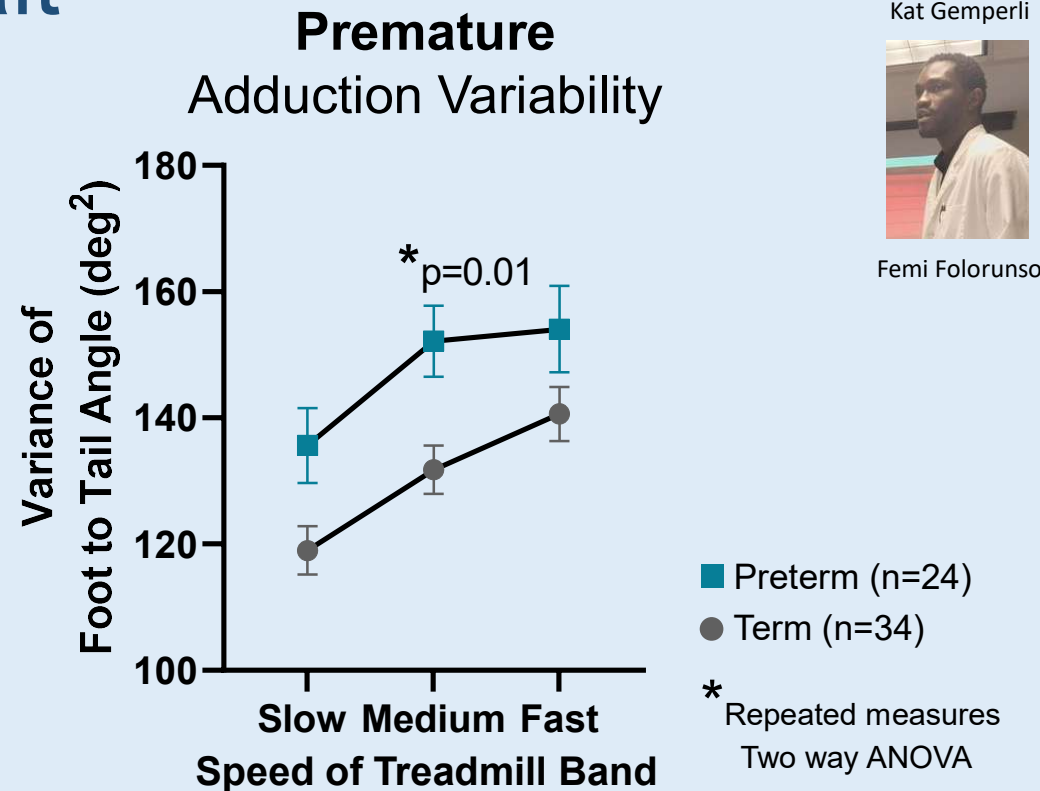
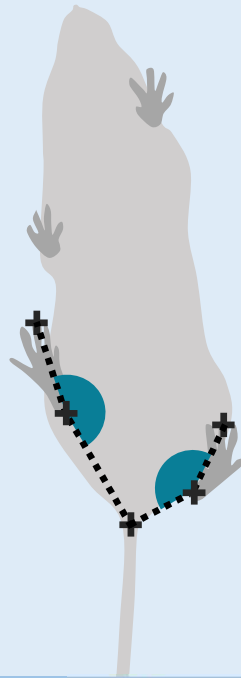


Kat Gemperli



Femi Folorunso

Assessing clinically-relevant dystonia features in mice - gait



Kat Gemperli



Femi Folorunso

Parvalbumin immunoreactivity reduced in the cortex in mice born preterm



Femi Folorunso

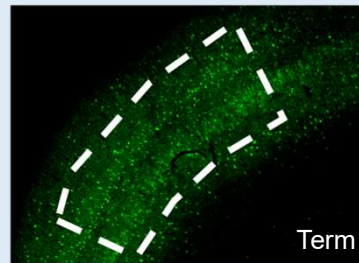
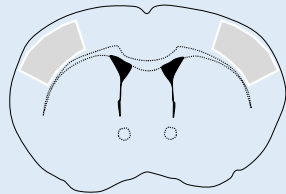


Rebecca Joshua

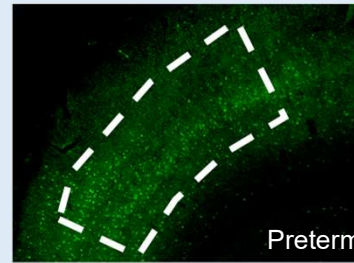


Rachel Rykowski

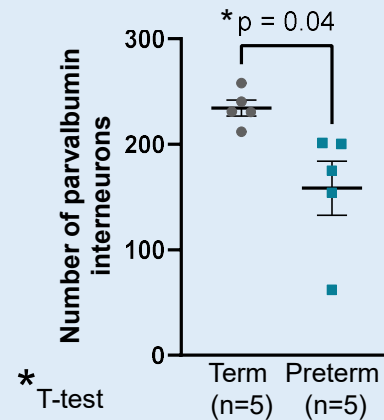
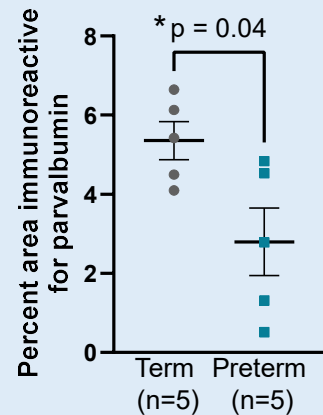
Sensorimotor cortex



Term



Preterm



* T-test

Parvalbumin immunoreactivity reduced in the cortex in mice born preterm



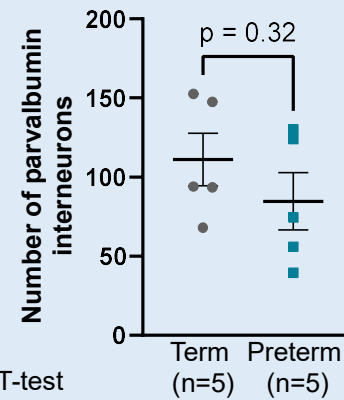
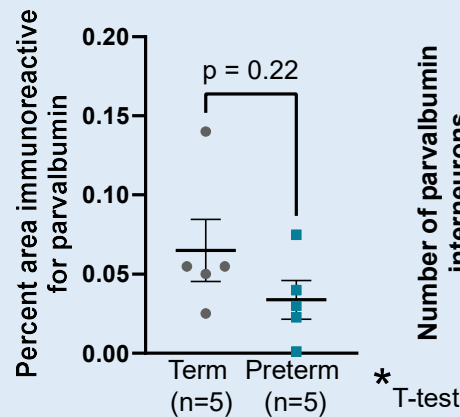
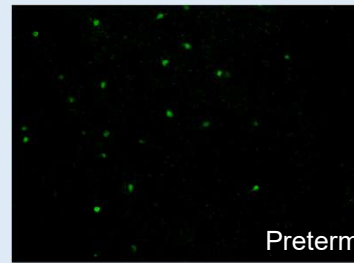
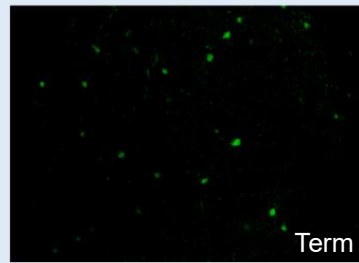
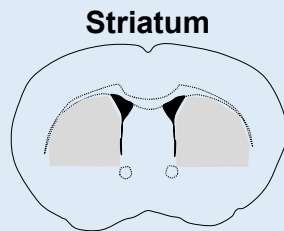
Femi Folorunso



Rebecca Joshua

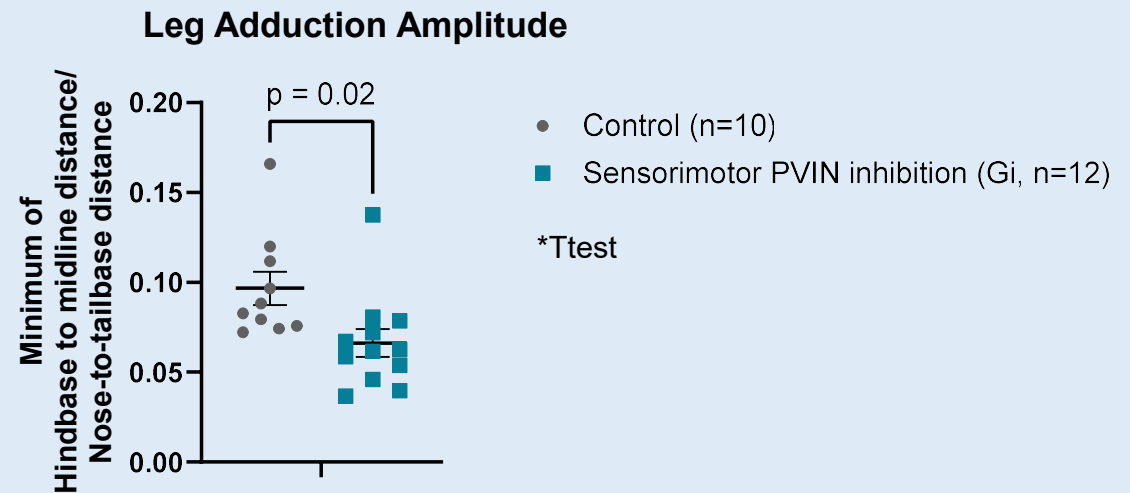
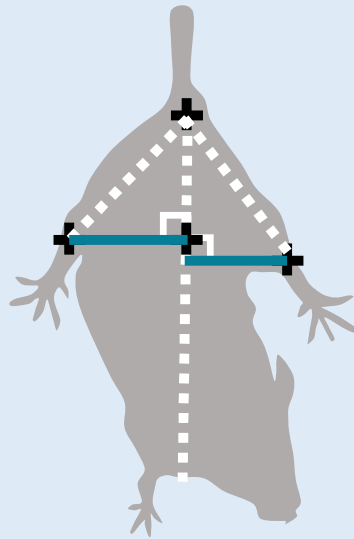


Rachel Rykowski

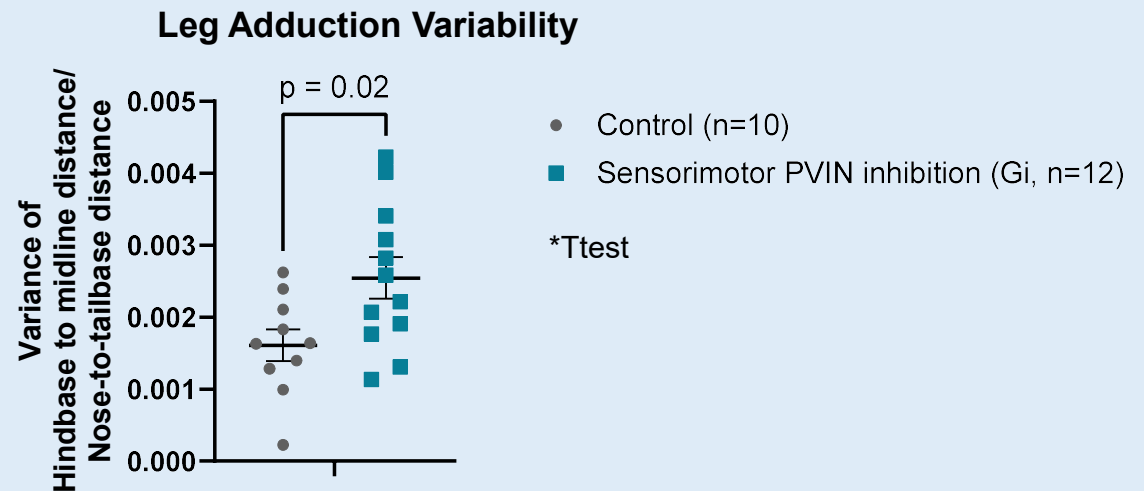
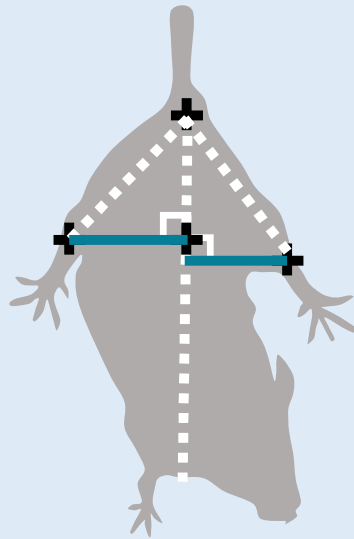


Does reduced activity of parvalbumin-positive GABAergic interneurons (PVINs) in the sensorimotor cortex **CAUSE** dystonia?

Does sensorimotor cortex PVIN inhibition cause dystonia?



Does sensorimotor cortex PVIN inhibition cause dystonia?



***Cortical dystonia of prematurity**

If measures that translate from
to do not exist,
we must **create** them



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